

Liquidity Traps, Prudential Policies, and International Spillovers

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This paper

- A model integrating macroprudential and monetary policy in a low-interest environment
- Research questions:
 - Are transmission channels of monetary and macroprudential policy interrelated?
 - What is the optimal joint policy?
 - Does macroprudential policy improve welfare?
- Model:
 - A small open New Keynesian economy with nominal rigidities
 - Occasionally binding zero lower bound constraint
 - Government sets the nominal interest rate and tax on capital flows

Preview of results

- Monetary policy spillovers operate mostly through expenditure switching
- Macroprudential policy spillovers operate mostly through intertemporal substitution
- In an open economy, macroprudential policy is a more precise instrument for influencing consumption-saving decision
- Using the two policies jointly reduces the likelihood of ZLB
- A risk of liquidity trap causes a trade-off between current output stabilization and reducing the probability of future recession

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Household preferences

- Household preferences are

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t z_t [U(c_t) - v(h_t)]$$

where z_t is a discount factor shock, h is labor supply

- CRRA utility with

$$c_t = \left[\omega \left(c_t^T \right)^{1-\frac{1}{\gamma}} + (1-\omega) \left(c_t^N \right)^{1-\frac{1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}$$

where

- ω is home bias
- γ is the elasticity of substitution

- Two one-period bonds
 - b^* is a real bond with a gross return R^* units of tradables
 - b is a nominal bond with a gross return R units of non-tradables
- Assume that the nominal bond can only be traded domestically
 - As households are identical, zero net supply in equilibrium
- Bonds are subject to a tax τ
 - $\tau > 0$ is tax on debt
 - $\tau < 0$ is tax on savings
- Can be interpreted as capital controls

Budget constraint

$$P_t^N c_t^N + P_t^T c_t^T + \frac{1}{1 + \tau_t} \left[\frac{b_{t+1}}{R_t} + P_t^T \frac{b_{t+1}^*}{R_t^*} \right] =$$
$$W_t h_t + \phi_t^N + P_t^T (y_t^T + T_t) + b_t + P_t^T b_t^*$$

where

- ϕ_t^N is profits
- y_t^T is endowment of tradable goods
- T_t is transfer from the government

- Firms produce non-tradable goods in a perfectly competitive market
- Labor is the only factor of production
- Profits are given by

$$\phi_t^N = P_t^N n_t^\alpha - W_t n_t$$

- Prices are perfectly rigid

$$P_t^N = \bar{P}^N$$

- Sets a nominal interest rate, $R_t \geq 1$
 - subject to zero lower bound
- Sets a tax on bond issuance τ (capital controls)
- The tax is rebated to households

$$T_t = -\frac{\tau_t}{1 + \tau_t} \left[\frac{b_{t+1}}{P_t^T R_t} + \frac{b_{t+1}^*}{R_t^*} \right]$$

- The law of one price holds, $P_t^T = e_t P_t^{T*}$
 - e_t is a nominal exchange rate
 - higher e_t is depreciation
- Risk-adjusted UIP holds

$$R_t^* = R_t \mathbb{E}_t \left[\Lambda_{t+1} \frac{e_t}{e_{t+1}} \frac{P_t^{T*}}{P_{t+1}^{T*}} \right]$$

where $\Lambda_{t+1} \equiv z_{t+1} u_T(c_{t+1}^T, c_{t+1}^N) / \mathbb{E}_t [z_{t+1} u_T(c_{t+1}^T, c_{t+1}^N)]$ is a stochastic discount factor

Equilibrium

Given an initial condition b_0^* , exogenous process $\{R_t^*, y_t^T, z_t\}_{t=0}^\infty$, a rigid price $\bar{P}^N$, and government policies $\{R_t, \tau_t\}_{t=0}^\infty$, an equilibrium is a stochastic sequence of prices $\{e_t, P_t^{T*}, W_t\}$ and allocations $\{c_t^T, c_t^N, b_{t+1}^*, n_t, h_t\}_{t=0}^\infty$ such that

- households optimize;
- firms choose hours to meet demand, $h^\alpha = c^N$;
- labor market clears and the domestic currency bond is in zero net supply;
- the government budget constraint is satisfied;
- the law of one price holds: $P_t^T = e_t P_t^{T*}$

First-best allocation

$$\max_{\{b_{t+1}^*, c_t^N, c_t^T\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t z_t \left[u(c_t^T, c_t^N) - v\left(\left(c_t^N\right)^{1/\alpha}\right) \right]$$

subject to the resource constraint for tradables

$$c_t^T = y_t^T + b_t^* - \frac{b_{t+1}^*}{R_t^*}$$

- Competitive equilibrium with flexible prices coincides with the first best
- With sticky prices but no ZLB, the government can replicate the first-best allocation

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Analytical Decomposition

- The aim is to understand how monetary and macroprudential policies are transmitted in open economies
- Study a one-period decline in the nominal interest rate at $t = 0$
- Impose simplifying assumptions
 - $\beta R^* = 1$
 - for $t \geq t_1$, the policy follows the first best allocation for the new steady state values
- Notice that CRRA preferences are additively separable over time
 - Can choose the allocation over time first and allocation over tradables/non-tradables second

Monetary policy effects on output

$$\frac{dy_0^N}{y_0^N} = \underbrace{-\beta\sigma(1-\tilde{\omega})\frac{dR_0}{R_0}}_{\text{Intertemporal Subst.}} + \underbrace{\tilde{\omega}\gamma\frac{de_0}{e_0}}_{\text{Expend. Switching}} - \underbrace{(1-\beta)(1-\tilde{\omega})[(1-\kappa)\sigma + \kappa\gamma]\frac{dR_0}{R_0}}_{\text{Aggregate Income}}$$

where

- $\tilde{\omega} = P_t^T c_t^T / (P_t^T c_t^T + P^N c_t^N)$, the share of tradables in expenditures
- $de_0/e_0 = -[1 - (1-\beta)(1-\kappa)(1-\sigma\gamma^{-1})] dR_0/R_0$, from UIP
- $\kappa \equiv \tilde{\omega} \left[1 + \frac{\alpha(1-\tilde{\omega})(\gamma-\sigma)}{\alpha\sigma + (1-\alpha+\phi)\gamma\sigma} \right]^{-1}$

Monetary policy effects on capital flows

$$\frac{db_1^*}{Pc} = \underbrace{\sigma(1 - \tilde{\omega}) \frac{dR_0}{R_0}}_{\text{Intertemporal Subst.}} - \underbrace{[(1 - \tilde{\omega})\sigma + \tilde{\omega}\gamma](1 - \tilde{\omega}) \frac{dR_0}{R_0}}_{\text{Aggregate Income}}$$

What is the net effect?

- If $\sigma > \gamma$, intertemporal substitution channel dominates and aggregate savings decrease
- If $\sigma < \gamma$, aggregate income channel dominates and aggregate savings increase
- If $\sigma = \gamma$, the aggregate savings are unchanged

Capital flows effect on monetary policy

- Compare the economy with closed and open capital accounts
- If $\sigma > \gamma$, output increases by more with an open capital account
 - Capital inflows allow shifting more consumption from future to present
- If $\sigma < \gamma$, output increases by more with a closed capital account
- If $\sigma = \gamma$, the output response is the same under the open and closed capital account

Macprudential policy effect on output

- Study a one-period change τ_0 in the tax on debt at $t = 0$
- The effect on output

$$\frac{dy_0^N}{y_0^N} = \underbrace{-\beta\sigma d\tau_0}_{\text{Intertemporal subst.}} + \underbrace{\tilde{\omega}\gamma \frac{de_0}{e_0}}_{\text{Expend. Switching}} - \underbrace{(1-\beta)(1-\kappa)\sigma d\tau_0}_{\text{Aggregate Income}}$$

where $\frac{de_0}{e_0} = -(1-\beta)\frac{1-\kappa}{1-\tilde{\omega}}\sigma\gamma^{-1}d\tau_0$

- The intertemporal substitution effect is stronger than the one of monetary policy
- The expenditure switching channel is weaker
 - Foreign savings next period increase, so the exchange rate has to appreciate to close the output gap
 - Current exchange rate also appreciates through the expected appreciation

Macroprudential policy effect on capital flows

$$\frac{db_1^*}{\mathcal{P}_C} = \underbrace{\sigma d\tau_0}_{\text{Intertemporal Subst.}} - \underbrace{(1 - \tilde{\omega})\sigma d\tau_0}_{\text{Aggregate Income}} = \sigma \tilde{\omega} d\tau_0$$

- Tax on debt unambiguously reduces capital inflows

The two policies operate through different channels

	$\sigma = 0.5$			$\sigma = 1$		
	$\gamma = 0.5$	$\gamma = 1$	$\gamma = 2$	$\gamma = 0.5$	$\gamma = 1$	$\gamma = 2$
Monetary Policy						
Change in y^N (pp)	0.50	0.62	0.91	0.89	1.00	1.27
Intertemporal subst.	0.37	0.36	0.34	0.74	0.71	0.69
Expend. switching	0.11	0.24	0.53	0.12	0.25	0.53
Aggregate income	0.02	0.02	0.04	0.03	0.04	0.05
Capital controls						
Change in y^N (pp)	0.50	0.50	0.50	1.01	1.01	1.01
Intertemporal subst.	0.48	0.48	0.47	0.97	0.96	0.96
Expend. switching	0.00	0.00	0.01	0.01	0.01	0.01
Aggregate income	0.02	0.02	0.02	0.03	0.04	0.04

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- Optimal macroprudential policy given arbitrary monetary policy
- Optimal joint monetary and macroprudential policy
- Monetary policy without capital controls

Macroprudential policy given an arbitrary monetary policy

- Given a sequence of $\{e_t\}$, the government chooses the state-contingent tax on debt $\{\tau_t\}$

$$\max_{\{b_{t+1}^*, c_t^N, c_t^T\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t z_t \left[u(c_t^T, c_t^N) - v\left((c_t^N)^{1/\alpha}\right) \right]$$

subject to

$$c_t^T = y_t^T + b_t^* - \frac{b_{t+1}^*}{R_t^*},$$
$$c_t^N = \left[\frac{1 - \omega}{\omega} \frac{p_t^{T*}}{\bar{p}^N} e_t \right]^\gamma c_t^T$$

Optimal policy intuition

- The aggregate demand externality:
 - While making savings decisions, households don't internalize the effect on the production of non-tradables
- The optimal capital controls correct the externality and depend on the labor wedge

$$\psi_t \equiv 1 - \frac{1}{\alpha h_t^{\alpha-1}} \frac{v'(h_t)}{u_N(c_t^T, c_t^N)}$$

- $\phi_t > 0$ means that a marginal value of employment is higher than the marginal cost from providing labor (a recession)

$$\tau_t = \frac{1}{\beta R_t^* \mathbb{E}_t \frac{z_{t+1}}{z_t} u_T(t+1)} \left\{ -\frac{1 - \tilde{\omega}_t}{\tilde{\omega}_t} u_T(t) \psi_t + \beta R_t^* \mathbb{E}_t \frac{z_{t+1}}{z_t} \left[\frac{1 - \tilde{\omega}_{t+1}}{\tilde{\omega}_{t+1}} u_T(t+1) \psi_{t+1} \right] \right\}$$

- No capital controls needed if $\psi_t = \psi_{t+1} = 0$
- If $\psi_t = 0$, then $\tau_t > 0$ when $\psi_{t+1} > 0$
 - Expected recession means it is optimal to tax capital inflows
- If $\psi_t > 0$ and $\psi_{t+1} > 0$, a trade-off between easing the recession tomorrow and reducing current demand

Joint monetary and macroprudential policies

$$V(b^*, s) = \max_{R, e, b', c^N, c^T} u(c^T, c^N) - v\left(\left(c^N\right)^{1/\alpha}\right) + \beta \mathbb{E}_{s'|s} \frac{z'}{z} V(b^{*'}, s')$$

subject to

$$c^T = y^T + b^* - \frac{b^{*'}}{R^*}$$

$$c^N = \left[\frac{1 - \omega}{\omega} \frac{p^{T*}}{\bar{p}^N} e \right]^\gamma c^T$$

$$R^* = R \mathbb{E}_{s'|s} \left[\Lambda(c^T(b^{*'}, s'), c^N(b^{*'}, s')) \frac{p^{T*}}{p^{T*'}} \frac{e}{\mathcal{E}(b^{*'}, s')} \right]$$

$$R \geq 1$$

Optimal policies intuition

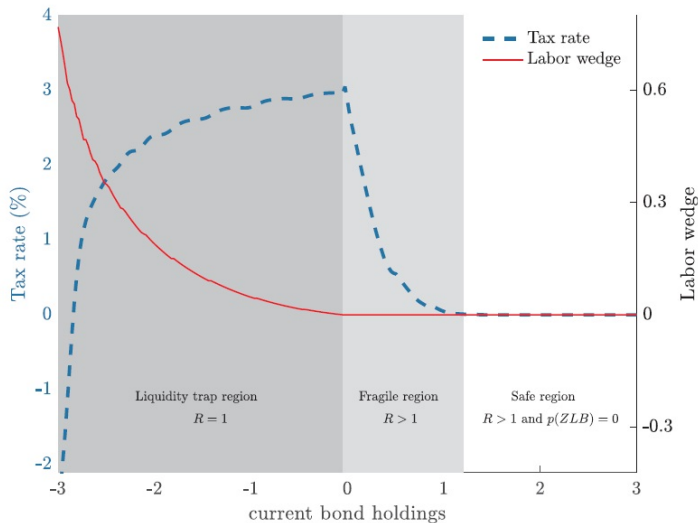
- If ZLB doesn't bind, can choose optimal monetary policy such that the labor wedge is fully stabilized
- If ZLB binds, the government is unable to depreciate the exchange rate and increase demand for non-tradables with monetary policy only
- Capital controls can change the intertemporal allocation of consumption instead

$$\tau = \frac{1}{\beta R^* \mathbb{E}_{s'|s} \frac{z'}{z} [u_T(c^{T'}, c^{N'})]} \left\{ -(1 + \Theta) \frac{\xi}{\gamma c^T} + \beta R^* \mathbb{E}_{s'|s} \frac{z'}{z} \left[\frac{\xi'}{\gamma c^{T'}} \right] \right\}$$

where $\Theta \equiv \gamma c^T \frac{\partial}{\partial b^{*'}} \mathbb{E}_{s'|s} \left[\frac{\Lambda(b^{*'}, s')}{\mathcal{E}(b^{*'}, s')} \frac{eP^{T*}}{P^{T*'}} \right]$ and ξ is the Lagrange multiplier on the ZLB constraint

- If ZLB doesn't bind currently but binds next period, $\tau > 0$, capital inflows are taxed
- If ZLB is binding now but not in the future, $\tau < 0$, capital outflows are taxed

The capital control is non-monotonic in the current level of bond holdings



Optimal monetary policy without capital control

$$V(b^*, s) = \max_{R, b^{*'}, c^N, c^T} u(c^T, c^N) - v\left((c^N)^{1/\alpha}\right) + \beta \mathbb{E}_{s'|s} \frac{z'}{z} V(b^{*'}, s')$$

subject to

$$c^T = y^T + b^* - \frac{b^{*'}}{R^*}$$

$$c^N = \left[\frac{1 - \omega}{\omega} \frac{p^{T*}}{\bar{P}N} e \right]^\gamma c^T$$

$$u_T(c^T, c^N) = \beta R^* \mathbb{E}_{s'|s} \left[\frac{z'}{z} u_T(c^T(b^{*'}, s'), c^N(b^{*'}, s')) \right]$$

$$R^* = R \mathbb{E}_{s'|s} \left[\Lambda(c^T(b^{*'}, s'), c^N(b^{*'}, s')) \frac{p^{T*}}{p^{T*'}} \frac{e}{\mathcal{E}(b^{*'}, s')} \right]$$

$$R \geq 1$$

(1)

Monetary policy without capital controls

- Should the government use monetary policy prudentially as a substitute for capital controls?
 - Stimulate precautionary savings, reduce the likelihood of liquidity traps
- Optimal monetary policy becomes

$$u_T(t)\psi_t = \frac{\tilde{\omega}_t(\sigma - \gamma)}{(1 - \tilde{\omega}_t)\sigma + \tilde{\omega}_t\gamma} \mathbb{E}_t \sum_{k=1}^{\infty} \beta^k \frac{z_{t+k}}{z_t} \left(\prod_{\ell=0}^{k-1} \frac{R_{t+\ell}^*}{1 + \bar{\Theta}_\ell} \right) \frac{\xi_{t+k}}{\gamma c_{t+k}^T}$$

$$\text{where } \bar{\Theta}_t \equiv \beta R_t^* \frac{1}{u_{TT}(t)} \mathbb{E}_t \frac{z_{t+1}}{z_t} \frac{\partial u_T(C^T(b_{t+1}^*, s_{t+1})), C^N(b_{t+1}^*, s_{t+1})}{\partial b_{t+1}^*}$$

Monetary policy without capital controls

- If $\sigma = \gamma$, savings don't respond to the nominal exchange rate, monetary policy cannot be used prudentially
- If $\sigma > \gamma$, increase the nominal interest rate to stimulate savings (leans against the wind)
 - Reduce the likelihood of future liquidity trap at the expense of recession today
- If $\sigma < \gamma$, decrease the nominal interest rate to stimulate savings (leans with the wind)
 - Reduce the likelihood of future liquidity trap at the expense of overheating today
- Notice that optimal monetary policy is used prudentially as long as ZLB may bind even in the distant future

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- United Kingdom data between 1980 and 2019
- The series of shocks are coming from
 - $\{y^T\}$ is a cyclical component of value added in the agriculture, mining, fishing and manufacturing from the World Development Indicators
 - $\{R^*\}$ is the US real interest rate
 - $\{z_t\}$ is a two-state Markov process targeting the average NFA/GDP
- $\sigma = 1$, $\gamma = \{0.5, 1, 2\}$
- Compare different policies in response to the series of shocks

Frequency and duration of liquidity traps

	Monetary Policy Only		Monetary & Macroprudential			
	Frequency	Duration	Frequency	Duration	mean(τ)	corr(R, τ)
$\gamma = 0.5$	3.4%	7.7	3.2%	11.0	0.2%	-0.1
$\gamma = 1$	3.8%	7.8	3.4%	9.6	0.2%	-0.3
$\gamma = 2$	4.3%	8.1	3.8%	9.0	0.2%	-0.5

- Capital controls reduce the likelihood of liquidity trap
- Lower external debt due to the taxes on inflows
- However, capital controls increase the average duration as the government taxes capital outflows and slow down the deleveraging process

Welfare costs of the ZLB and unemployment rate

	Monetary Policy Only		Monetary & Macroprudential	
	Welfare costs	Unemployment*	Welfare costs	Unemployment*
$\gamma = 0.5$	0.47%	7.80%	0.08%	1.44%
$\gamma = 1$	0.40%	5.99%	0.09%	1.46%
$\gamma = 2$	0.45%	5.10%	0.12%	1.65%

- The welfare cost of the ZLB is the compensating consumption variations that equalize the expected utility of a household in this economy and the expected utility in the efficient allocation
- Unemployment is the average conditional on a liquidity trap

Conclusion

- In an open economy, intertemporal substitution is not the only channel of monetary policy
- Capital controls operate mostly through intertemporal substitution
- Capital controls can play the prudential role against ZLB better than monetary policy

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Description	Parameter Value	Source/Target
Intertemporal elasticity	$\sigma = 1$	Standard value
Technology	$\alpha = 1$	Standard value
Frisch elasticity parameter	$\phi = 3$	Gali and Monacelli (2005)
Weight on tradables in CES	$\omega = 0.25$	Share of tradable output = 24%
Discount factor	$\beta = 0.99$	Standard value
Discount factor shocks	$g^H = 0.006$	Average NFA-GDP ratio = -17.4%
Transition prob. g^L to g^H	$P(g^L g^H) = 0.20$	3 liquidity traps every century
Transition prob. g^H to g^L	$P(g^L g^L) = 0.39$	2 years duration of liquidity traps
Intratemporal elasticity	$\gamma \in \{0.5, 1, 2\}$	

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